

CostAllocator Pro - Complete Project Documentation

Finmark.ai × Denave | P&L AutoTrack Suite

1. PROJECT OVERVIEW

Project Name: CostAllocator Pro
Brand: Finmark.ai × Denave
Type: Enterprise Financial Intelligence Platform
Status: Live (Phase 1 Complete)
Version: 1.0.0

2. LIVE URLS

Service	URL
Frontend	https://costallocator-pro.vercel.app
Backend API	https://costallocator-api-3skb.onrender.com
API Documentation (Swagger)	https://costallocator-api-3skb.onrender.com/docs
GitHub Repository	https://github.com/hvs-finmarkai/costallocator-pro

3. CREDENTIALS & API KEYS

3.1 Application Login

Field	Value
Email	admin@finmark.ai
Password	Admin@123456
Role	super_admin

3.2 Neon PostgreSQL Database

Field	Value
Provider	Neon (Free Tier)
Project ID	hidden-glitter-59720153
Region	aws-ap-southeast-1
Host	ep-purple-recipe-az37wddy.c-3.ap-southeast-1.aws.neon.tech
Database	neondb
User	neondb_owner

Password	npg_ym6MlecX0fut
Connection String	postgresql://neondb_owner: npg_ym6MlecX0fut@ep-purple-recipe-az37wddy.c-3.ap-southeast-1.aws.neon.tech /neondb?sslmode=require
Dashboard	https://console.neon.tech

3.3 Vercel (Frontend Hosting)

Field	Value
Team	fin-markai
Project	costallocator-pro
Framework	Next.js 16
Domain	costallocator-pro.vercel.app

3.4 Render (Backend Hosting)

Field	Value
Service Name	costallocator-api
Runtime	Python 3.11
Region	Oregon
Plan	Free
Dashboard	https://dashboard.render.com

3.5 GitHub

Field	Value
Organization	hvs-finmarkai
Repository	costallocator-pro
Branch	master
URL	https://github.com/hvs-finmarkai/costallocator-pro

3.6 Environment Variables (Backend - Render)

Key	Value
DATABASE_URL	postgresql://neondb_owner: npg_ym6MlecX0fut@ep-purple-recipe-az37wddy.c-3.ap-southeast-1.aws.neon.tech /neondb?sslmode=require
SECRET_KEY	(auto-generated by Render)
CORS_ORIGINS	[" https://costallocator-pro.vercel.app ", " http://localhost:3000 "]

DEBUG	true
PYTHON_VERSION	3.11.9

3.7 Environment Variables (Frontend - Vercel)

Key	Value
NEXT_PUBLIC_API_URL	https://costalocator-api-3skb.onrender.com

4. TECH STACK

4.1 Frontend

Technology	Version	Purpose
Next.js	16.2.10	React Framework
React	19	UI Library
TypeScript	5.x	Type Safety
TailwindCSS	4.x	Styling
shadcn/ui	Latest	UI Components
Zustand	Latest	State Management
React Query	Latest	API Data Fetching
React Hook Form	Latest	Form Handling
Zod	Latest	Validation
Recharts	Latest	Charts
Lucide React	Latest	Icons
next-themes	Latest	Dark/Light Mode
Framer Motion	Latest	Animations

4.2 Backend

Technology	Version	Purpose
FastAPI	0.115.0	API Framework
SQLAlchemy	2.0.35	ORM
Alembic	1.13.3	Database Migrations
Pydantic	2.9.2	Data Validation
bcrypt	4.0.1	Password Hashing

python-jose	3.3.0	JWT Tokens
psycopg2-binary	2.9.9	PostgreSQL Driver
uvicorn	0.30.6	ASGI Server
redis	5.1.1	Caching (planned)
httpx	0.27.2	HTTP Client

4.3 Database

Technology	Purpose
PostgreSQL 17 (Neon)	Primary Database
Redis (planned)	Caching & Queues

4.4 Deployment

Service	Platform
Frontend	Vercel (Free)
Backend	Render (Free)
Database	Neon (Free)
CI/CD	GitHub → Auto-deploy

5. PROJECT STRUCTURE

```
costallocator-pro/
├── frontend/                                # Next.js 16 Application
│   ├── src/
│   │   ├── app/
│   │   │   ├── (auth)/
│   │   │   │   ├── login/page.tsx
│   │   │   │   ├── register/page.tsx
│   │   │   │   ├── forgot-password/page.tsx
│   │   │   │   └── layout.tsx
│   │   │   ├── (dashboard)/
│   │   │   │   ├── executive/page.tsx
│   │   │   │   ├── pnl/
│   │   │   │   │   ├── page.tsx
│   │   │   │   │   ├── revenue/page.tsx
│   │   │   │   │   ├── costs/page.tsx
│   │   │   │   │   └── margins/page.tsx
│   │   │   │   ├── allocation/
│   │   │   │   │   ├── page.tsx
│   │   │   │   │   ├── cost-centers/page.tsx
│   │   │   │   │   └── rules/page.tsx
```

```

| | | | | └─ simulation/page.tsx
| | | | |   └─ pricing/page.tsx
| | | | | └─ forecast/
| | | | |   └─ page.tsx
| | | | |     └─ ai/page.tsx
| | | | | └─ alerts/page.tsx
| | | | | └─ reports/page.tsx
| | | | | └─ admin/
| | | | |   └─ page.tsx
| | | | |   └─ users/page.tsx
| | | | |   └─ roles/page.tsx
| | | | |   └─ audit/page.tsx
| | | | |     └─ settings/page.tsx
| | | | |   └─ layout.tsx
| | | | └─ globals.css
| | | | └─ layout.tsx
| | | |   └─ page.tsx
| | | └─ components/
| | |   └─ charts/
| | |   └─ layout/
| | |   └─ shared/
| | |     └─ ui/
| | └─ config/
| | └─ hooks/
| | └─ services/
| | └─ store/
| |   └─ types/
| └─ .env.production
| └─ package.json
|   └─ vercel.json
|
└─ backend/                                # FastAPI Application
    └─ app/
        └─ api/
            └─ v1/
                └─ endpoints/
                    └─ auth.py
                    └─ users.py
                    └─ revenues.py
                    └─ expenses.py
                    └─ pnl.py
                    └─ __init__.py
                └─ core/
                    └─ config.py
                    └─ security.py
                └─ db/
                    └─ base.py
                    └─ session.py
                └─ models/
                    └─ user.py
                    └─ organization.py
                    └─ role.py

```

```
| | | | └─ revenue.py
| | | | └─ expense.py
| | | | └─ cost_center.py
| | | | └─ budget.py
| | | | └─ exchange_rate.py
| | | | └─ audit_log.py
| | | | └─ refresh_token.py
| | | └─ schemas/
| | | | └─ auth.py
| | | | └─ user.py
| | | | └─ revenue.py
| | | | └─ expense.py
| | | | └─ pnl.py
| | | └─ services/
| | | | └─ audit.py
| | └─ main.py
| └─ alembic/
| | └─ versions/
| | | └─ 001_initial_schema.py
| | | └─ 002_add_pnl_tables.py
| | └─ env.py
| └─ migrate.py
| └─ Procfile
| └─ requirements.txt
| └─ runtime.txt
| └─ Dockerfile
| └─ .env.example
|
└─ docker-compose.yml
└─ render.yaml
└─ .gitignore
└─ README.md
```

6. DATABASE SCHEMA

6.1 Existing Tables

Table	Description
organizations	Company/tenant data
users	User accounts with roles
roles	Custom role definitions
permissions	Granular permissions
role_permissions	Role-permission mapping
refresh_tokens	JWT refresh token storage
audit_logs	All system activity logs

revenues	Revenue entries
expenses	Cost/expense entries
cost_centers	Shared cost centers
budgets	Budget allocations
exchange_rates	Currency conversion rates

6.2 Users Table

Column	Type	Description
id	UUID	Primary key
email	VARCHAR(255)	Unique, indexed
hashed_password	VARCHAR(255)	bcrypt hash
first_name	VARCHAR(100)	
last_name	VARCHAR(100)	
role	ENUM	super_admin, cfo, coo, finance_controller, account_manager, hr, operations, viewer
avatar	VARCHAR(500)	Profile image URL
is_active	BOOLEAN	Account status
organization_id	UUID	FK → organizations
created_at	TIMESTAMP	Auto
updated_at	TIMESTAMP	Auto

6.3 Revenues Table

Column	Type	Description
id	UUID	Primary key
organization_id	UUID	FK → organizations
client_name	VARCHAR(255)	
project_name	VARCHAR(255)	
business_unit	VARCHAR(100)	
geography	VARCHAR(100)	
revenue_type	ENUM	recurring, one_time, project_based, milestone, time_material
status	ENUM	draft, recognized, deferred, invoiced, collected

amount	NUMERIC(15,2)	
currency	VARCHAR(10)	Default: INR
amount_inr	NUMERIC(15,2)	Converted amount
period_start	DATE	
period_end	DATE	
invoice_number	VARCHAR(100)	
description	TEXT	
source	VARCHAR(50)	manual, sap, salesforce, etc.

6.4 Expenses Table

Column	Type	Description
id	UUID	Primary key
organization_id	UUID	FK → organizations
cost_center_id	UUID	FK → cost_centers
client_name	VARCHAR(255)	
project_name	VARCHAR(255)	
business_unit	VARCHAR(100)	
category	ENUM	employee, vendor, infrastructure, travel, software, marketing, operational, overhead, other
expense_type	ENUM	direct, indirect, shared
description	TEXT	
amount	NUMERIC(15,2)	
currency	VARCHAR(10)	
amount_inr	NUMERIC(15,2)	
period	DATE	
vendor_name	VARCHAR(255)	
is_recurring	BOOLEAN	
source	VARCHAR(50)	

6.5 Audit Logs Table

Column	Type	Description
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id	UUID	Primary key
user_id	UUID	FK → users
organization_id	UUID	FK → organizations
action	VARCHAR(50)	login, create, update, delete, export
resource	VARCHAR(100)	auth, user, revenue, expense, etc.
resource_id	VARCHAR(255)	ID of affected record
details	JSONB	Additional context
ip_address	VARCHAR(45)	Client IP
user_agent	TEXT	Browser info
created_at	TIMESTAMP	

7. API ENDPOINTS

7.1 Authentication

Method	Endpoint	Description
POST	/api/v1/auth/login	Login with email/password
POST	/api/v1/auth/register	Register new organization
GET	/api/v1/auth/me	Get current user
POST	/api/v1/auth/refresh	Refresh access token
POST	/api/v1/auth/logout	Revoke all sessions
POST	/api/v1/auth/forgot-password	Request password reset

7.2 Users

Method	Endpoint	Description
GET	/api/v1/users	List users (with search, pagination)
POST	/api/v1/users	Create user (admin only)
GET	/api/v1/users/{id}	Get user details
PATCH	/api/v1/users/{id}	Update user
DELETE	/api/v1/users/{id}	Delete user (admin only)

7.3 Revenue

Method	Endpoint	Description
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GET	/api/v1/revenues	List revenues (with filters)
POST	/api/v1/revenues	Create revenue entry
GET	/api/v1/revenues/stats	Revenue statistics
GET	/api/v1/revenues/{id}	Get revenue details
PATCH	/api/v1/revenues/{id}	Update revenue
DELETE	/api/v1/revenues/{id}	Delete revenue

7.4 Expenses

Method	Endpoint	Description
GET	/api/v1/expenses	List expenses (with filters)
POST	/api/v1/expenses	Create expense entry
GET	/api/v1/expenses/stats	Expense statistics
GET	/api/v1/expenses/{id}	Get expense details
PATCH	/api/v1/expenses/{id}	Update expense
DELETE	/api/v1/expenses/{id}	Delete expense

7.5 P&L

Method	Endpoint	Description
GET	/api/v1/pnl/summary	Full P&L summary (gross/contribution/net margin)
GET	/api/v1/pnl/margin-trend	Monthly margin trend

7.6 Health

Method	Endpoint	Description
GET	/health	Service health check
GET	/	Service info

8. FRONTEND PAGES (22 Pages)

#	Page	Route	Status
1	Login	/login	✔ Functional
2	Register	/register	✔ Functional
3	Forgot Password	/forgot-password	✔ UI Ready
4	Executive Dashboard	/executive	✔ UI Ready

5	P&L Overview	/pnl	✓ UI Ready
6	Revenue	/pnl/revenue	✓ UI Ready
7	Costs	/pnl/costs	✓ UI Ready
8	Margins	/pnl/margins	✓ UI Ready
9	CostAllocator Overview	/allocation	✓ UI Ready
10	Cost Centers	/allocation/cost-centers	✓ UI Ready
11	Allocation Rules	/allocation/rules	✓ UI Ready
12	Simulation	/allocation/simulation	✓ UI Ready
13	Pricing Advisor	/allocation/pricing	✓ UI Ready
14	Forecast	/forecast	✓ UI Ready
15	AI Insights	/forecast/ai	✓ UI Ready
16	Alerts	/alerts	✓ UI Ready
17	Reports	/reports	✓ UI Ready
18	Admin Overview	/admin	✓ UI Ready
19	Users	/admin/users	✓ Functional (CRUD)
20	Roles	/admin/roles	✓ UI Ready
21	Audit Logs	/admin/audit	✓ UI Ready
22	Settings	/admin/settings	✓ UI Ready

9. SECURITY FEATURES

Feature	Status
JWT Access Tokens (30 min expiry)	✓
Refresh Token Rotation	✓
Token Blacklist (DB stored)	✓
bcrypt Password Hashing (12 rounds)	✓
RBAC Role Enforcement	✓
Organization Isolation (multi-tenant)	✓
Audit Logging (all mutations)	✓
CORS Configuration	✓
Input Validation (Pydantic)	✓

Auth Guard (frontend)	
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10. GIT COMMITS HISTORY

```
6affd70 fix: sidebar now responds to theme toggle
bac5cc3 fix: dark mode theme toggle not working
73b0b68 fix: force black font color on all input fields
194fa36 fix: input text color - black in light mode, white in dark mode
c66302d fix: sidebar collapsed state - show only icons, proper alignment
dc36155 feat: make Admin Users page fully functional (Add + Delete)
0acd716 fix: remove DropdownMenu from Header to fix Base UI error #31
a989f63 fix: fix navigation links and add all missing pages
f8f9313 fix: align frontend types with backend snake_case API response
221882a fix: handle null user state in Header and AuthGuard hydration
df4b1c8 fix: add frontend .env.production file (Render API URL)
2243a2f fix: include .env.production with Render API URL for build-time env
fdb457 feat: add auth guard - redirect to login if not authenticated
2ca7877 fix: use create_all(checkfirst=True) instead of alembic for deploy
84a082d feat: add CostAllocator, Alerts, and Admin pages
9337347 fix: stamp migration 001 before upgrade to handle existing tables
24edc8e feat: add P&L frontend pages (Revenue, Costs, Margins)
47429ce feat: add Module 1 - P&L AutoTrack Core (Revenue, Cost, Margin engines)
106760a fix: use alembic stamp head for existing DB tables
64ec262 fix: add DATABASE_URL directly in render.yaml
be89ffb fix: resolve circular import in backend startup
bee6515 refactor: harden backend for enterprise production on Render
577bf1e feat: add dark/light theme toggle with next-themes
ac3ce51 fix: resolve Vercel serverless deployment issues
70a519e feat: add deployment configs for Vercel + Render
f5b0f39 feat: initialize CostAllocator Pro - Phase 1 infrastructure
```

11. HOW TO RUN LOCALLY

11.1 Frontend

```
cd frontend
npm install
npm run dev
# Opens at http://localhost:3000
```

11.2 Backend

```
cd backend
python -m venv venv
venv\Scripts\activate          # Windows
pip install -r requirements.txt
# Create .env file with DATABASE_URL and SECRET_KEY
```

```
uvicorn app.main:app --reload
# Opens at http://localhost:8000
```

11.3 Database (Docker)

```
docker-compose up -d
# PostgreSQL at localhost:5432
# Redis at localhost:6379
```

12. DEPLOYMENT PROCESS

Frontend (Vercel)

1. Push to GitHub master branch
2. Run: `cd frontend && npx vercel --yes --prod`
3. Auto-deploys at `costallocator-pro.vercel.app`

Backend (Render)

1. Push to GitHub master branch
 2. Render auto-deploys from GitHub
 3. Build command: `pip install -r requirements.txt && python migrate.py`
 4. Start command: `uvicorn app.main:app --host 0.0.0.0 --port $PORT`
-

13. WHAT'S BUILT vs WHAT'S REMAINING

Built (Phase 1) — ~40%

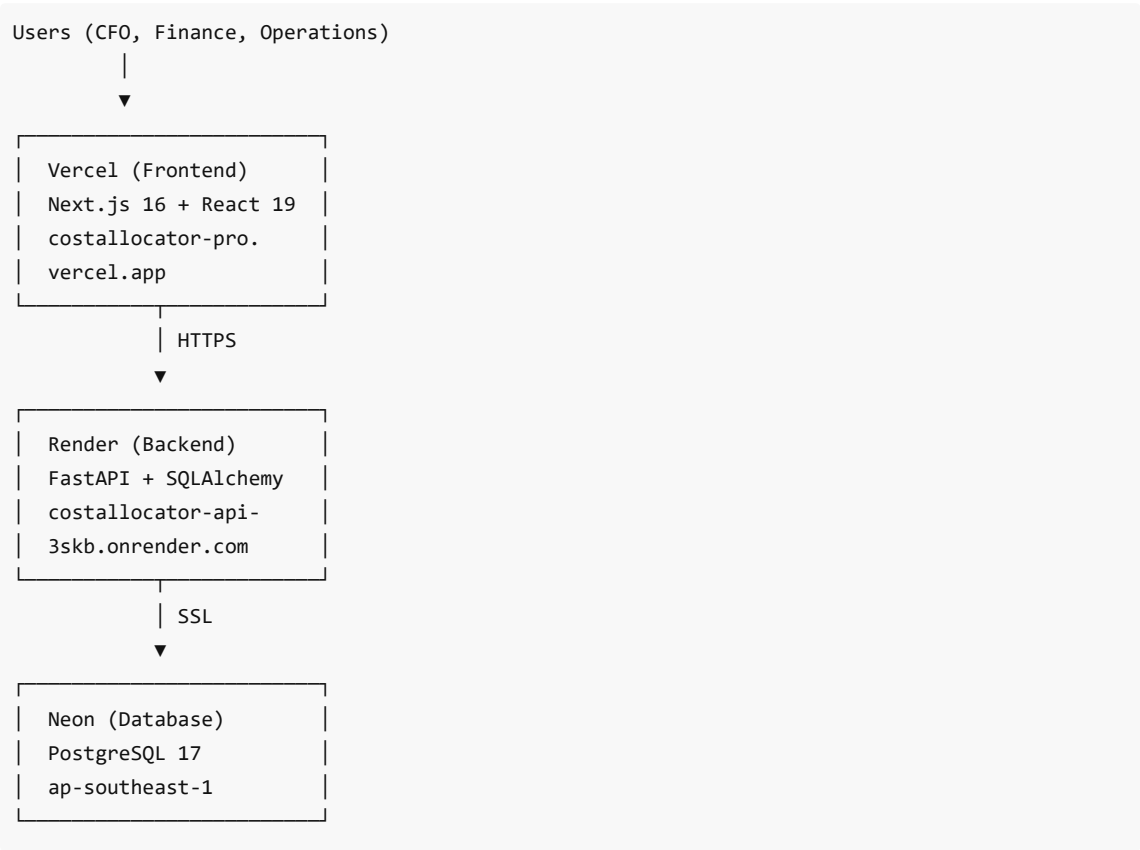
-  Authentication (Login, Register, JWT, Refresh, Logout)
-  RBAC (Roles, Permissions, Enforcement)
-  Organization Isolation (Multi-tenant)
-  Audit Logging
-  User Management (CRUD)
-  Executive Dashboard (UI)
-  P&L Management (Revenue, Costs, Margins pages)
-  CostAllocator Pro (Cost Centers, Rules, Simulation, Pricing pages)
-  Alerts, Reports, Forecast pages
-  Dark/Light Theme Toggle
-  Revenue & Expense APIs (CRUD + Stats)
-  P&L Summary & Margin Trend APIs

Remaining (Phases 2-6) — ~60%

-  Revenue Recognition Engine
-  Workforce/Employee Management
-  Cost Allocation Engine (actual calculations)
-  Forecast/AI Predictions
-  Background Jobs (Celery)
-  Email Notifications
-  Report Generation (PowerPoint, PDF, Excel)
-  SAP/Salesforce/Tally Integration

- ❌ WebSocket Real-time Updates
- ❌ Data Import (CSV Upload)
- ❌ Multi-Factor Authentication
- ❌ SSO (Google, Microsoft, Azure AD)
- ❌ Monitoring (Grafana, Sentry)
- ❌ CI/CD Pipeline (GitHub Actions)

14. ARCHITECTURE DIAGRAM



15. CONTACT & OWNERSHIP

Field	Value
Organization	Finmark.ai
Partner	Denave
Neon Account	harsh.singh@finmark.ai
GitHub Org	hvs-finmarkai

Document Generated: July 19, 2026

Version: 1.0.0

Classification: Internal/Confidential